PNS

Bayes theorem:

Statement:

Let $E_1, E_2, \ldots, E_n$ be n mutually exclusive events defined on sample space (S) with $P(E_i) \neq 0$ where $i = 1, 2, 3, \ldots, n$.
For any arbitrary event A, which is subset of S such that $P(A) > 0$, then the prob of occurence of $P(E_i)$ given that A has already occured is denoted by $P(E_i/A)$ and given as,

$$\therefore \boxed{P(E_i/A) = \frac{P(E_i) \cdot P(A/E_i)}{\sum_{i=1}^{n} P(E_i) \cdot P(A/E_i)}}$$

where, $P(E_i/A)$ = posterior probability
$P(E_i)$ = Prior probability
$P(A/E_i)$ = Conditional probability.

Proof: From def^n of conditional probability.

$$P(E_i/A) = \frac{P(E_i \cap A)}{P(A)}$$

$$P(E_i/A) = \frac{P(E_i) \cdot P(A/E_i)}{P(A)} \longrightarrow (1)$$

Here, $E_1, E_2, \ldots E_n$ are mutually exclusive events defined in S. Then S can be written as,

$$S = E_1 \cup E_2 \cup E_3 \cup \ldots \cup E_n$$

Also event A is also a subset of S,

$$A = A \cap S$$
$$= A \cap [E_1 \cup E_2 \cup \ldots \cup E_n]$$
$$= (A \cap E_1) \cup (A \cap E_2) \cup (A \cap E_3) \cup \ldots \cup (A \cap E_n) \longrightarrow (2)$$

Since $E_1, \ldots E_n$ are mutually exclusive events, $(A \cap E_1)$, $(A \cap E_2) \ldots (A \cap E_n)$ are also mutually exclusive events.

Now,
taking prob. on both sides,

$$P(A) = P[(A\cap E_1)\cup(A\cap E_2)\ldots]$$

$$P(A) = P[(A\cap E_1)+(A\cap E_2)+\ldots+(A\cap E_n)]$$

$$P(A) = \sum_{i=1}^{n}(A\cap E_i)$$

$$P(A) = \sum_{i=1}^{n} P(E_i).P(A/E_i) \longrightarrow ③$$

Put ③ in ①,

$$P(E_i/A) = \frac{P(E_i).P(A/E_i)}{\sum_{i=1}^{n} P(E_i).P(A/E_i)}$$

Q) A coin is tossed until a head appears. What is the expectation of the number of tosses required to get first head?

Sol^n,

Let H and T denote occurrence of head & tail when tossing a coin respectively.
Let X be the R.V. that represents the no. of tosses required to get first head.
Then the head occurs in following ways:

Outcomes	No. of tosses required (X)	$P(X=x)$	$xP(X=x)$
H	1	1/2	1/2
TH	2	1/4	2/4
TTH	3	1/8	3/8
TTTH	4	1/16	4/16
⋮	⋮	⋮	⋮
			$\sum_{x=1}^{\infty} \frac{1}{2}+\frac{2}{4}+\frac{3}{8}+\frac{4}{16}+\ldots$

So,

$$S = \sum(x) = \frac{1}{2} + \frac{2}{4} + \frac{3}{8} + \frac{4}{16} + \dots \longrightarrow (1)$$

Dividing (1) by (2) on both sides

$$\frac{S}{2} = \frac{1}{4} + \frac{2}{8} + \frac{3}{16} + \frac{4}{32} + \dots \longrightarrow (2)$$

Subtract (2) from (1)

$$S - \frac{S}{2} = \frac{1}{2} + \left(\frac{2}{4} - \frac{1}{4}\right) + \left(\frac{3}{8} - \frac{2}{8}\right) + \left(\frac{4}{16} - \frac{3}{16}\right) + \dots$$

$$\frac{S}{2} = \frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16} + \frac{1}{32} + \dots$$

$$\frac{S}{2} = \frac{1/2}{1 - 1/2} \Rightarrow 1 \qquad \left\{ \quad S = \frac{a}{1-r}, \quad r < 1 \right.$$

geometric mean (G.M)

So, $S = 2$

$\therefore E(X) = 2$

Binomial distibution : ~~Imp~~

$$X \sim B(n, p)$$

$$P(X = x) = {}^{n}C_{x}\, p^{x} q^{n-x} \quad ; \qquad x = 0, 1, 2, \dots, n.$$

X = no. of success

n = no. of trials

p = prob. of success $= 1 - q$

q = prob of failure $= 1 - p$

Note : $n \leq 20$

$p \geq 0.05$

Properties: $E(X) = np$

$V(x) = npq$, $SD = \sqrt{V(x)} = \sqrt{npq}$

$V(x) < E(x)$

0, 1, 2

4, 3, 2

$P(X \geq 2) = \sum_{x=2}^{4} (0.5)^{x} (0.5)^{4-x} = 11/16$

Expected frequency $= N \times P(X \geq 2)$

$= 8000 \times \frac{11}{16}$

$= 5500$

$P(X \geq 3)$

$n = 6$ $\quad P(X \geq 2)$

$p = 0.2$

$q = 0.8$

$P(X \geq 2) = \sum_{x=2}^{6} {}^{6}C_{x} (0.2)^{x} (0.8)^{6-x}$

Poisson distribution:

$n \to \infty$ (very big) ; $p \to 0$ (very small)

$X \sim P(\lambda)$

$np = \lambda$ is finite , λ = parameter (avg no. of success)

$$P(X = x) = \frac{e^{-\lambda} \cdot \lambda^{x}}{x!} \quad ; \quad x = 0, 1, 2, \ldots$$

Properties : $E(X) = \lambda$, $\simeq$ $V(x) = \lambda$

$n > 20$, $p < 0.05$

$$\lambda = np$$

$\lambda = 2.8$

$$P(X=0) = \frac{e^{2.8}\ 2.8^{0}}{0!}$$

$$P(X \geq 3) = 1 - P(X < 3)$$

Normal distribution

$X \sim N(\mu, \sigma^2)$

$Z = \frac{x-\mu}{\sigma}$, the variate Z is known as standard normal variate (SND) and it's prob. distribution is standard normal distribution.

X = normal variate

Standard normal distribution:
The normal distribution with mean 0 and variance 1. $X \sim N(0,1)$

$SD = \sigma$
$V = \sigma^2$

$\sigma \rightarrow SD$, $V(x) = \sigma^2$

$SD = \sigma = \sqrt{V(x)}$

Rectangular (uniform)

$$f(x) = \frac{1}{b-a} \; ; \quad a \le x \le b \; , \quad X \sim U(a,b)$$

$$\text{Mean} = \frac{a+b}{2} \; , \quad \text{Variance} = \frac{(b-a)^2}{12}$$

Exponential distribution

$$f(x) = \lambda e^{-\lambda x} \; , \quad x \ge 0 \; , \quad \lambda > 0 \; , \quad X \sim \text{Exp}(\lambda)$$

$$E(X) = \frac{1}{\lambda} \; , \quad V(X) = \frac{1}{\lambda^2}$$

Karl Pearson correlation coefficient.

$$r = \frac{\operatorname{cov}(X,Y)}{\sqrt{V(X)}\,\sqrt{V(Y)}} \Rightarrow \frac{n\Sigma XY - \Sigma X \Sigma Y}{\sqrt{n\Sigma X^2 - (\Sigma X)^2} \cdot \sqrt{n\Sigma Y^2 - (\Sigma Y)^2}}$$

if r lies bet^n 0 and ± 0.499 low degree of positive or negative correlation

if r lies bet^n ± 0.5 and ± 0.699, average degree

if r lies bet^n ± 0.7 and ± 0.999, high degree

Spearmann's rank correlation coefficient.

Case I: ranks not repeated

$$r_s = 1 - \frac{6\Sigma d^2}{n(n^2-1)}$$

$$d = R_1 - R_2$$

Case II: Ranks are repeated

$$r_s = 1 - \frac{6\left[\Sigma d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m^2-1)}{12} + \dots\right]}{n(n^2-1)}$$

m = no. of times rank is repeated.

Regression eqn & regression lines:

1) Regression eqn of Y on X.

$$\boxed{Y = a + bX} \longrightarrow (1)$$

where

Y = dependent variable

X = Independent variable

a = constant (Y-intercept)

$b = b_{yx}$ = slope (regression coefficient of Y on X)

We solve by using ordinary least square method

$$\Sigma Y = na + b\Sigma X \longrightarrow (2)$$

$$\Sigma XY = a\Sigma X + b\Sigma X^2 \longrightarrow (3)$$

Solving (2) & (3), we get a and b.

Standard error of estimate (S_E)

$$S_E = \sqrt{\frac{\Sigma Y^2 - a\Sigma Y - b\Sigma XY}{n-2}}$$

Coefficient of determination (R^2)
($r^2 = R^2$)

$$R^2 = \frac{a\Sigma Y + b\Sigma XY - n(\bar{Y})^2}{\Sigma Y^2 - n(\bar{Y})^2}$$

Criteria of good estimator:

→ An estimator is said to be a good estimator of pop^n parameter if it satisfies:

i) Unbiasedness:
An estimator, t is said to be unbiased estimator of pop^n parameter θ if $E(t) = \theta$

ii) Consistency:
An estimator t is said to be consistent estimator of pop^n parameter θ if $t \xrightarrow{P} \theta$ as $n \to \infty$

iii) Efficiency:
Consider two consistent estimator t_1 & t_2 of pop^n parameter θ. Then t_1 is said to be more efficient than t_2 if $V(t_1) < V(t_2)$

iv) Sufficiency:
An estimator t is said to be sufficient estimator of θ if it contains all information of θ.

Confidence interval: $100(1-\alpha)\%$

Confidence interval for popⁿ mean(μ):

Case I: $n > 30$ → normal distribution.
$(\sigma = s)$

Formula: The $100(1-\alpha)\%$ confidence interval for popⁿ mean is given by,

$$\boxed{\bar{X} \pm Z_{\frac{\alpha}{2}} \times SE(\bar{X})}$$

$SE(\bar{X})$ = standard error of sample mean

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \quad \text{(for infinite pop}^n\text{)}$$

$$SE(\bar{X}) = \frac{\sigma}{\sqrt{n}} \cdot \sqrt{\frac{N-n}{N-1}} \quad \text{(for finite pop}^n\text{)}$$

$Z_{\frac{\alpha}{2}}$ = critical value SND at α level of significance for two tailed test.

Case 2: $n \le 30$
student's t-distribution

$$\bar{X} \pm t_{\frac{\alpha}{2}, n-1} \times SE(\bar{X})$$

$df = n-1$

$$SE(\bar{X}) = \frac{s}{\sqrt{n}}.$$

If sample SD is given, it is known as biased SD.
So we use $SE(\bar{X}) = \frac{s}{\sqrt{n-1}}$

We use this

$$S^2 = \frac{1}{n-1} \Sigma (X - \bar{X})^2$$

$$= \frac{1}{n-1} \left(\Sigma X^2 - \frac{(\Sigma X)^2}{n} \right)$$

$$S = \sqrt{\frac{1}{n-1} \left[\Sigma X^2 - \frac{(\Sigma X)^2}{n} \right.}$$

Confidence interval for population proportion:

$$P = \frac{X}{N} \quad (X \rightarrow \text{no. of success}), \quad Q = 1 - P$$

(proportion of success in population)

$$p = \frac{x}{n}, \quad q = 1 - p$$

(proportion of success in sample size)

$$p \pm Z_{\frac{\alpha}{2}} \, SE(p)$$

where, $p = \frac{x}{n}$

$SE(p) = \sqrt{\frac{pq}{n}}$ (for infinite popⁿ)

$SE(p) = \sqrt{\frac{pq}{n}} \cdot \sqrt{\frac{N-n}{N-1}}$ (for finite popⁿ)

$Z_{\frac{\alpha}{2}}$ = Critical value of SND at α level of significance for two tailed test.

Confidence interval for difference of two pop^n means ($\mu_1 - \mu_2$)

Case I: For $n_1 > 30$ and $n_2 > 30$

$$\left[(\bar{X}_1 - \bar{X}_2) \pm z_{\frac{\alpha}{2}} SE(\bar{X}_1 - \bar{X}_2)\right]$$

$$SE(\bar{X}_1 - \bar{X}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} \qquad (\sigma \simeq s)$$

Case II: For $n_1 \leq 30$, $n_2 \leq 30$

$$\left[(\bar{X}_1 - \bar{X}_2) \pm t_{\frac{\alpha}{2}, n_1+n_2-2} \cdot SE(\bar{X}_1 - \bar{X}_2)\right]$$

$$SE(\bar{X}_1 - \bar{X}_2) = \sqrt{S_p^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$S_p^2 = \frac{1}{n_1+n_2-2}\left(\Sigma X_1^2 - \frac{(\Sigma X_1)^2}{n_1} + \Sigma X_2^2 - \frac{(\Sigma X_2)^2}{n_2}\right)$$

Confidence interval for difference of two pop^n proportions

$$\left[(P_1 - P_2) \pm z_{\frac{\alpha}{2}} SE(P_1 - P_2)\right]$$

$$P_1 = \frac{x_1}{n_1}, \quad P_2 = \frac{x_2}{n_2}$$

$$SE(P_1 - P_2) = \sqrt{\hat{p}\hat{q}\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

$$\hat{p} = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2} = \frac{x_1 + x_2}{n_1 + n_2}$$